

Practical 3

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Electricity and Electronics EBN 111

Semester 1, 2024

Practical Information

Maximum marks:	200	Full marks:	200
Duration of practical:	180 min	Open/Closed book:	Open
Total number of pages (this page included)		14	

Lecturers: Prof X. Ye, Mr J. Thiselton, Mr L. Staphorst**Assistant Lecturers:** S. Twala, L. Hurter, J Masela, J. Nepembe**Overview of Practicals**

The engineering process to build any circuit consists of three parts:

- **Calculations**

A circuit design is proposed. The circuit is analyzed using fundamental circuit theory, e.g., Ohm's Law, Kirchoff's Laws, Nodal analysis, Mesh analysis etc. Calculations are done by hand to determine specific voltages and/or currents in the circuit. This is the first step to ensure the proposed circuit design is correct.

- **Simulation**

The proposed circuit is simulated using software such as LTspice. The simulation includes non-ideal effects which are not included in the calculations. The simulation results should coincide closely with the calculations done previously. The simulation allows us to analyze the circuit for a wide variety of circumstances. The simulations verify that the proposed circuit will perform its intended function.

- **Laboratory/Practical**

The proposed circuit is built using physical components. The hardware is connected to an actual power source such as a battery. Laboratory equipment can be used to measure the voltages or currents determined in the calculation and simulation steps above. The measurements should coincide closely with the results in the *Calculation* step. It will not be exact as there are tolerances in the physical components which are not included in the *Calculations*. The measurements should also be similar to the simulations, but will not be exactly the same since not all non-ideal characteristics are included in the simulation.

Practicals Process

The EBN 111 Practical will follow the engineering process discussed above. As shown in the figure below, a complete Practical consists of a **Pre-Practical**, a **Demonstration**, and a **Laboratory** part. The Laboratory part is also often referred to as a 'Practical session'.

1. Section 1: Pre-Practical - Calculations and Simulation [60 marks]

- Analyse the circuit diagrams in the practical guide using the theory covered in EBN 111. *Calculate* the various unknowns.
- *Simulate* the circuits using LTSpice and compare the results to your calculations.
- Download the Excel spreadsheet for your Pre-Practical on AMS and fill in your answers. Submit your Excel spreadsheet on AMS before the relevant deadline.
- You will have to work through the guideline documentation to familiarise yourself with the circuit components and software that are required for the pre-practical.

Pre-Practicals are completed at home before a Practical session.

2. Section 2: Practical Demonstration [20 marks]

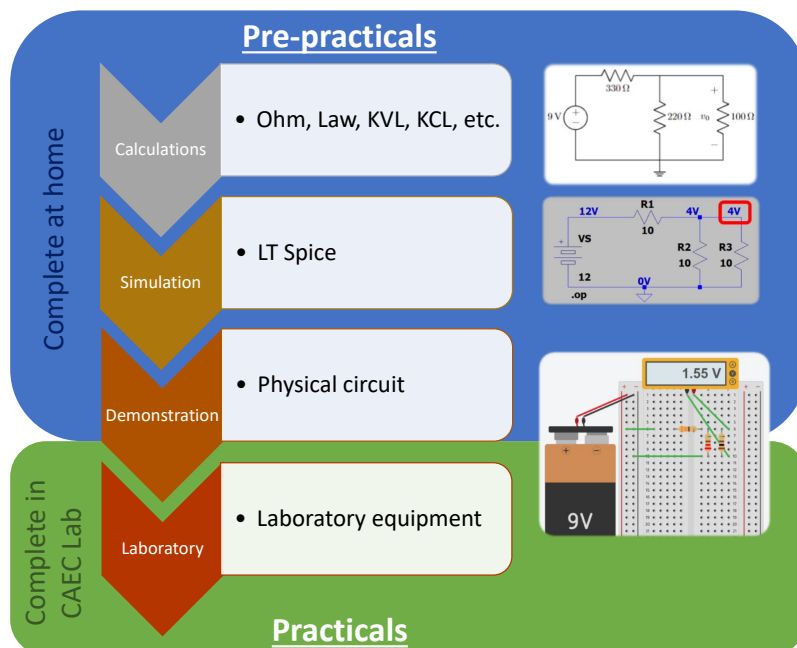
- You will be given a circuit similar to a circuit you analyzed and simulated to build on a breadboard using the component kit provided for EBN 111. You will demonstrate this physical circuit upon entering the laboratory. We will award a mark for the circuit.

3. Section 3: Practical Measurements [120 marks]

- Use laboratory equipment to measure voltages and currents as specified in the practical guide for the circuit you built. Compare it to your Pre-Practical results.
- Download the Excel spreadsheet for your Practical on AMS and fill in your answers. Submit your Excel spreadsheet on AMS before the end of the practical session.

Section 3 is completed in the CAEC Lab during a 3-hour Practical session.

All measurements must be made and submitted during the Practical session.



Assessment evaluation, submission rules and time-line

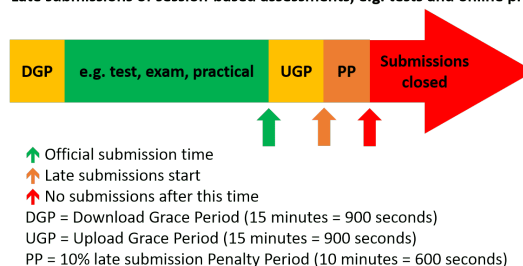
The following guidelines apply when filling in answers on an Excel spreadsheet.

- The Excel spreadsheet for a Pre-Practical or Practical is found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully when filling in your answers.
- Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write irrational numbers. Use decimal points and NOT decimal commas.
- Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also, refer to the example list of units on the Excel spreadsheet.)

After filling in your answers in an Excel spreadsheet for a Pre-Practical or Practical, you need to upload the Excel spreadsheet to AMS for that assessment to be marked. Each Pre-practical and Practical has a start time and an end-time as indicated on AMS. The following rules and timeline apply for all assessments for EBN 111 as illustrated by the figure below:

1. The Download Grace Period (DGP) is a 15 minute period before the official start of the assessment for students to download all documents related to the assessment and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the assessment.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the assessment.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn111.2024@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers during this period without incurring any penalties. No email submissions will be accepted during the UGP.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheet during this period, the student incurs a 10% penalty to their final assessment mark.
8. No late submissions are accepted after the PP.

Late submissions of session-based assessments, e.g. tests and online practicals



By uploading an Excel spreadsheet to AMS, you declare the following:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment.

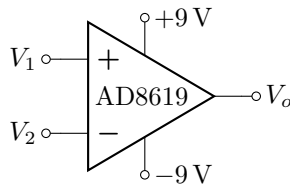
Section 1: Pre-Practical - Calculations and Simulation

Questions 1 to 6 [12]

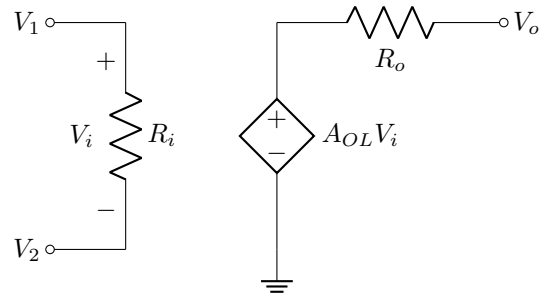
Given: V_{in}

The non-ideal op-amp shown in **Circuit 1** can be modelled using the equivalent op-amp model shown in **Circuit 2** for theoretical calculations and simulations.

Circuit 1

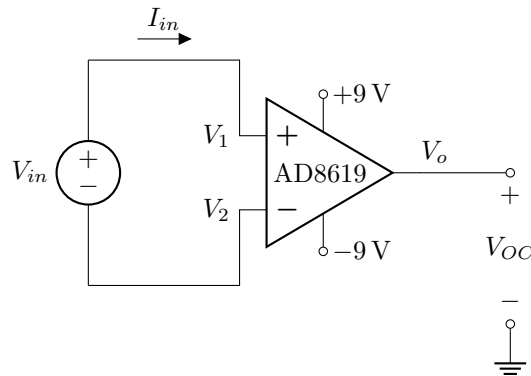


Circuit 2



The parameters required for the equivalent op-amp circuit in **Circuit 2** can be found experimentally by performing a few tests on the non-ideal AD8619 op-amp (this can be assumed to be the LTspice equivalent for the $\mu A741$ op-amp that you will use in Section 3) in **Circuit 1** as described below.

Circuit 3

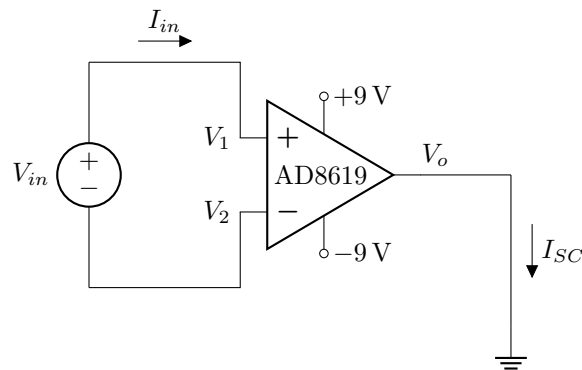


Build **Circuit 3** in LTspice and perform a **DC operating point** simulation. Use the simulated values to complete the questions below to find the parameters of **Circuit 2**.

- | | |
|-----------|--|
| 01 | Measure the current I_{in} in simulation. [2] |
| 02 | Determine the input resistance R_i of the op-amp using I_{in} and Ohm's law. [2] |
| 03 | Measure the open-circuit (Thevenin) output voltage V_{OC} in simulation. [2] |
| 04 | Determine the open-loop gain A_{OL} of the op-amp. [2] |

Build **Circuit 4** in LTspice and perform a **DC operating point** simulation. Use the simulated values to complete the questions below to find the parameters of **Circuit 2**.

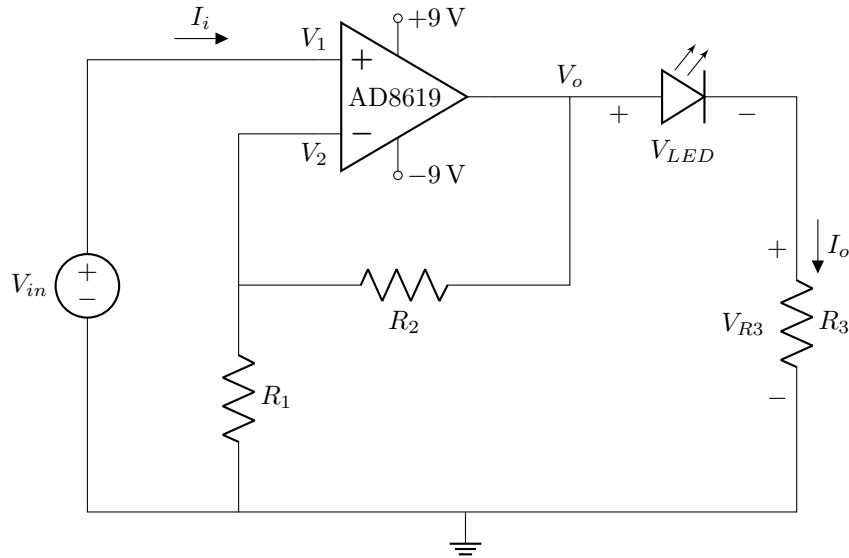
Circuit 4



05 Measure the short-circuit (Norton) output current I_{SC} in simulation. [2]

06 Determine the output resistance R_o of the op-amp. [2]

Questions 7 to 23 [22]

Given: $R_1, R_2, R_3, V_{in}, V_{LED}$ **Circuit 5**Assume an ideal op-amp in **Circuit 5**. Complete the questions below through theoretical analysis.

- | | |
|-----------|--|
| 07 | Calculate V_1 . [1] |
| 08 | Calculate V_2 . [1] |
| 09 | Calculate I_i . [1] |
| 10 | Calculate V_o . [1] |
| 11 | Calculate the overall gain $A_T = V_o/V_{in}$. [1] |
| 12 | Calculate V_{R3} . [1] |
| 13 | Calculate I_o . [1] |
| 14 | Calculate the power P_{LED} dissipated by the LED. [1] |

Build **Circuit 5** in LTspice using the AD8619 opamp and the QTLP690C LED. Perform a **DC operating point** simulation and complete the questions below.

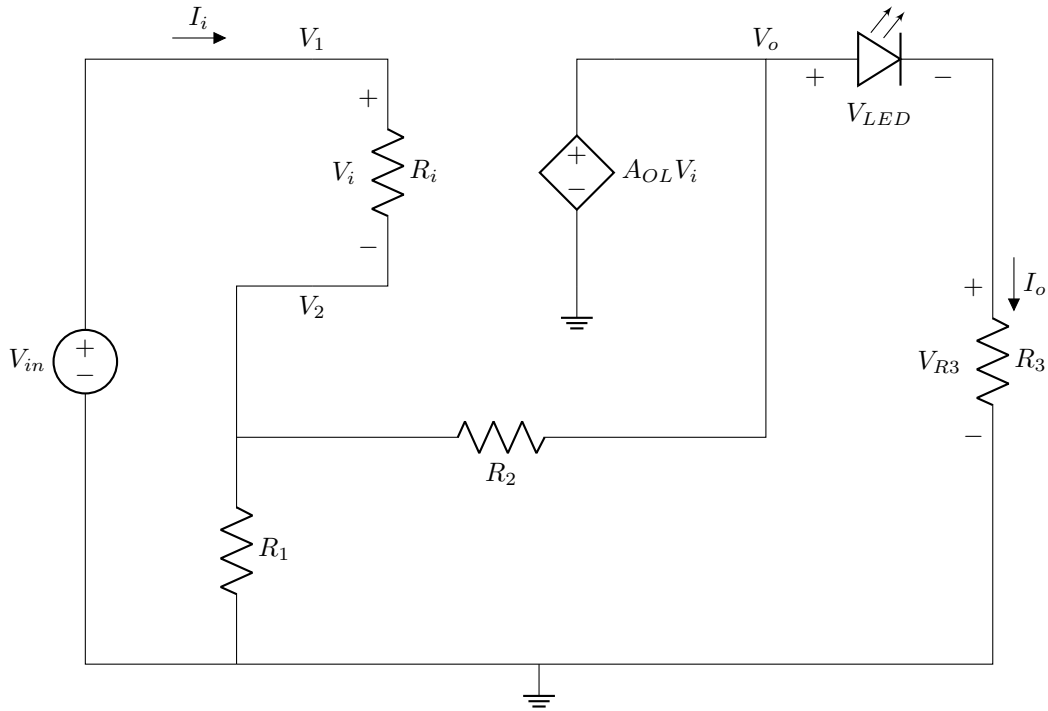
- | | |
|-----------|--|
| 15 | Measure V_1 in simulation. [1] |
| 16 | Measure V_2 in simulation. [1] |
| 17 | Measure I_i in simulation. [1] |
| 18 | Measure V_o in simulation. [2] |
| 19 | Determine the overall gain $A_T = V_o/V_{in}$ in simulation. [1] |
| 20 | Measure V_{R3} in simulation. [2] |
| 21 | Measure I_o in simulation. [2] |
| 22 | Measure V_{LED} in simulation. [2] |
| 23 | Measure the power P_{LED} dissipated by the LED in simulation. [2] |

Questions 24 to 42 [26]

Given: $R_1, R_2, R_3, V_{in}, V_{LED}, R_i, A_{OL}$

Important: The values for R_1, R_2, R_3, V_{in} , and V_{LED} are the same as for Q7 to Q23 above.

If we do not assume the op-amp is ideal, we can replace the op-amp in **Circuit 5** with an equivalent circuit which models the non-ideal characteristics. The equivalent circuit for the op-amp is shown **Circuit 6** where we assume $R_o = 0\Omega$. We will use this circuit for subsequent theoretical calculations and simulations.

Circuit 6

Complete the questions below through theoretical analysis of **Circuit 6**.

24	Calculate V_1 . [1]
25	Calculate V_2 . [1]
26	Calculate V_i . [1]
27	Calculate I_i . [1]
28	Calculate V_o . [1]
29	Calculate the overall gain $A_T = V_o/V_i$. [1]
30	Calculate V_{R3} . [1]
31	Calculate I_o . [1]
32	Calculate the power P_{LED} dissipated by the LED. [1]
Build Circuit 6 in LTspice using a dependent voltage source and the QTLP690C LED. Perform a DC operating point simulation and complete the questions below.	
33	Measure V_1 in simulation. [1]
34	Measure V_2 in simulation. [1]
35	Measure V_i in simulation. [1]

36	Measure I_i in simulation. [2]
37	Measure V_o in simulation. [2]
38	Determine the overall gain $A_T = V_o/V_i$ in simulation. [2]
39	Measure V_{R3} in simulation. [2]
40	Measure I_o in simulation. [2]
41	Measure V_{LED} in simulation. [2]
42	Measure the power P_{LED} dissipated by the LED in simulation. [2]

Total Section 1: [60]

Section 2: Practical Demonstration

Question 43 [20]

Given: R_2 , R_3 , R_5

Build **Circuit 7** shown below on a breadboard using the components provided in the component kit. A 9 V battery can be used to power the circuit. This circuit will be used for the practical session. It is advised to read through the next section and build the circuit in a way that the changes required for the Practical will be easy to implement.

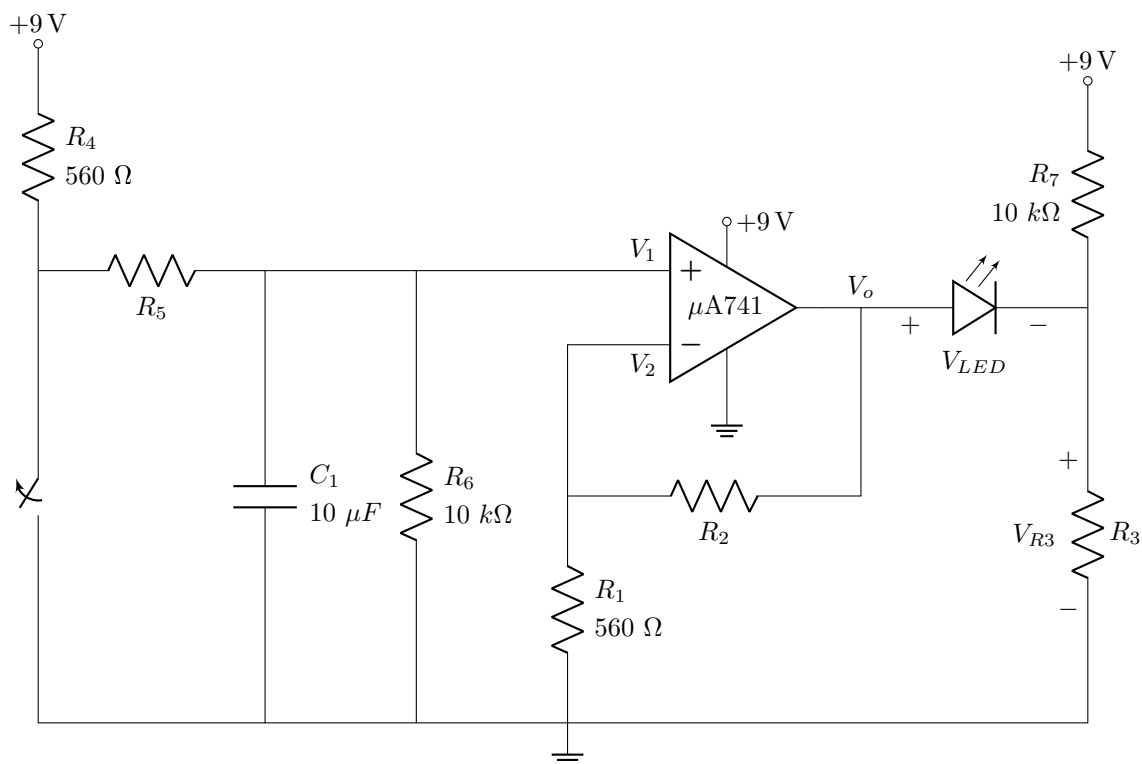
After building the circuit, you can test it by pressing the button to close the connection. The brightness of the LED should gradually increase or decrease when the button is pressed or released.

When entering the lab for your allocated practical session, the circuit will be evaluated by an Assistant Lecturer.

Please ensure that your circuit is prepared before the practical session.

You will not be allowed to build the circuit during the practical session.

Circuit 7



43	Circuit evaluation [20]
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Total Section 2: [20]

Section 3: Practical Measurements

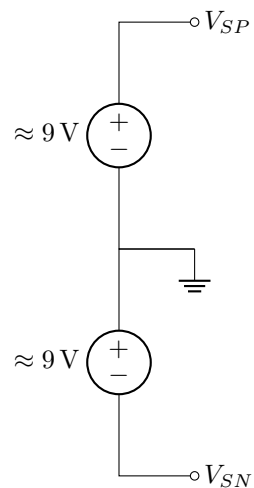
Questions 44 to 45 [4]

Set up the laboratory power supply to supply both positive and negative 9 V DC. This is illustrated in **Circuit 8**. As shown, you will need to make use of two outputs of the laboratory power supply. Use the multimeter to measure the positive supply voltage V_{SP} and verify that it is between 8.9 V and 9.1 V. Similarly, measure the negative supply voltage V_{SN} and verify that it is between -9.1 V and -8.9 V. Enter the measured supply voltages below.

**The measured supply voltage V_S will be used in the marking of subsequent questions.
Take care to ensure that this step is done correctly.**

Once the supply voltages are set, they should not be changed under any circumstances!

Circuit 8



44 V_{SP} [2]

45 V_{SN} [2]

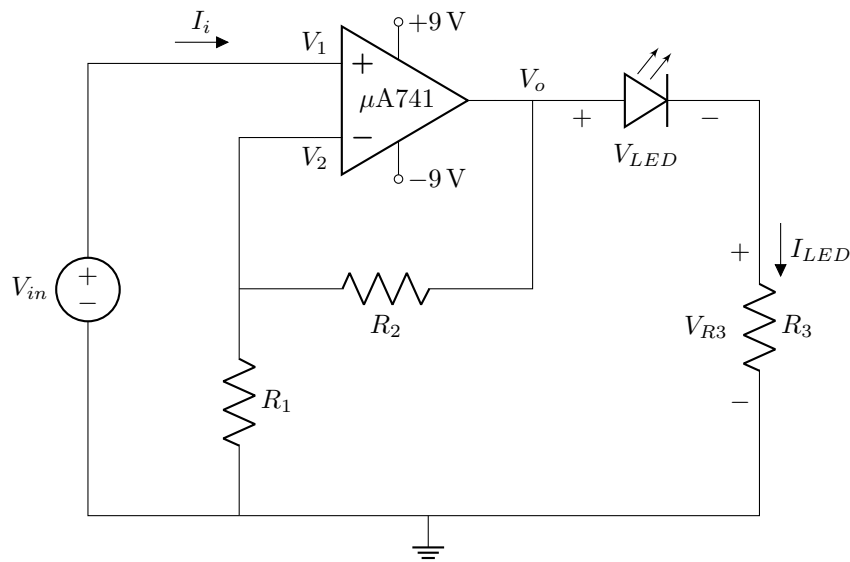
Questions 46 to 54 [54]Given: R_1 , R_2 , R_3 , V_{in}

Important: The values for R_1 , R_2 , R_3 and V_{in} should be same as used for Q7 to Q23 above and will be used for the rest of the Practical.

Remove the capacitor, switch, and resistors R_4 , R_5 , R_6 , and R_7 from **Circuit 7**. Replace resistors R_1 , R_2 and R_3 with resistors with the required R_1 , R_2 and R_3 values from Q7 to Q23. Remove the battery from the circuit and connect the laboratory power supply to provide the op-amp with both positive (V_{SP}) and negative (V_{SN}) voltage. The power supply was set to $\pm 9\text{ V}$ DC in a previous question and the actual output voltage was measured as V_{SP} and V_{SN} .

The circuit should resemble **Circuit 9** shown below.

Use the multimeter to measure the voltages/currents specified below. Remember that current must be measured in series. However, current can be calculated by measuring the voltage across a component first and then using Ohm's law.

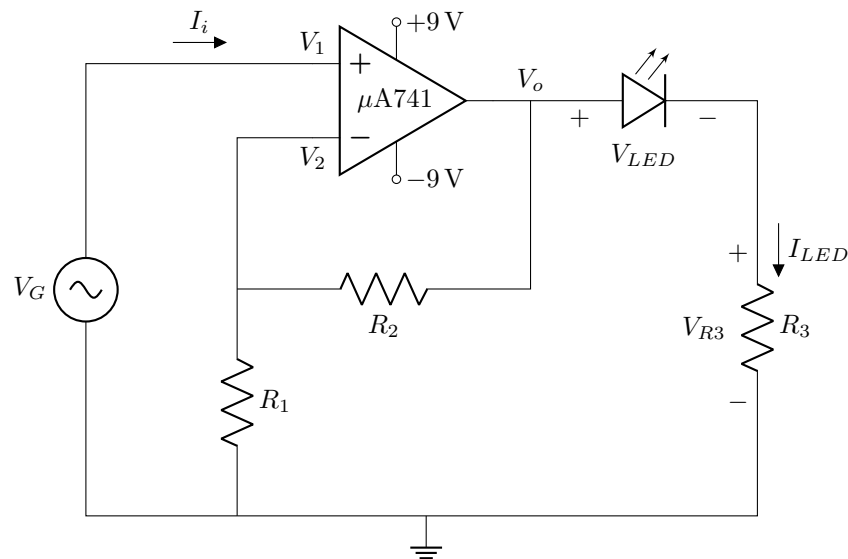
Circuit 9

46	Measure V_1 . [8]
47	Measure V_2 . [8]
48	Measure I_i . [8]
49	Measure V_o . [8]
50	Determine the overall gain $A_T = V_o/V_{in}$. [2]
51	Measure V_{R3} . [8]
52	Measure I_{LED} . [2]
53	Measure V_{LED} . [8]
54	Measure the power P_{LED} dissipated by the LED. [2]

Questions 55 to 61 [32]	
Given: f	
Set up the signal generator to supply a sinusoidal signal V_G with a peak-to-peak amplitude of 1 V, a 0 V offset, and a frequency of f Hz. Use the oscilloscope to measure the signal V_G. Adjust the V/div setting of the oscilloscope such that the signal amplitude fills the screen vertically without exceeding the bounds of the screen. Adjust the t/div setting of the oscilloscope until at least one full period of the sinusoidal signal is in view. Answer the questions below.	
55	What is your V/div setting on the oscilloscope? [1]
56	What is your t/div setting on the oscilloscope? [1]
57	What is the number of vertical (voltage) divisions from the signal minimum to its maximum? [6]
58	What is the Peak-to-peak amplitude V_{pp} of V_G ? [6]
59	What is the number of horizontal (time) divisions in one signal period? [6]
60	What is the Period T of V_G ? [6]
61	What is the frequency f_T of V_G found using the period T ? [6]

Questions 62 to 66 [30]

Apply the signal V_G from the signal generator as set in a previous question to the input of the op-amp as shown in **Circuit 10**.

Circuit 10

Use the oscilloscope to measure the output voltage V_o and answer the questions below.

62	Measure the peak-to-peak voltage V_{pp} of V_o . [8]
63	Measure the period T of V_o . [8]
64	Measure the frequency f_T of V_o using the period T . [2]

Replace R_2 in **Circuit 10** with an LDR and remove the LED as shown in **Circuit 11** below. Vary how much light falls on the LDR and observe how the output signal changes with varying levels of illumination. Answer the questions below.

Circuit 11

65 Does the gain of the circuit increase (write 1), decrease (write 2), or remain unchanged (write 3) when the illumination increases (i.e., more light falls on the LDR)? [6]

66 Does the frequency of the circuit increase (write 1), decrease (write 2), or remain unchanged (write 3) when the illumination increases? [6]

Total Section 3: [120]
Grand Total : [200]